

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 5

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Create a multi-process system where a master and a worker both attempt to increment a shared counter. You will observe why shared memory provides raw speed but lacks the safety of message queues.

To do this, create a header file `mym.h`, which contains a struct with an integer called `counter`. As we discussed in class, apply the `volatile` keyword to the counter to prevent the compiler from caching its value in a CPU register.

Write a program `master.c` that opens a shared memory object for the counter and increments it 1,000,000 times. Write another program `worker.c` that connects to the shared memory object the master created and increments the counter 1,000,000 times. After running both programs concurrently, what is the final value of the counter? Explain the result.

2. Implement the same system using message queues. Use the Linux `time` command to measure the execution of both implementations. How does the performance of the shared memory-based system compare to the message queue version?

- **Welcome: Sean**

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